

Undergraduate Project Criteria

The following rubric outlines the relevant content of the project for each section. The relative weight of each criterion is outlined. For information on the grading scale, see the attached document.

A common structure of quantitative research projects is: Introduction, Data Exploration, Data Analysis, Conclusion and Discussion.

Generally, you want to use the introduction to clearly state your research question, define the relevant economic phenomena you'll be investigating, review the possible relationships you might expect to find and the rationale behind those hypotheses. You may also preview your results. From there, you want to connect these theoretical ideas to something tangible—your data.

You should start this by explaining what data you have access to for your analysis, where it comes from, why it exists, and what information it contains. After reviewing the relevant meta-information for your data, then you can concretely connect it to your research question by identifying the key variables you'll use to explore your relationship of interest. When you do so, you'll need to explain any transformations you make to these variables to make them more suitable for your analysis (and the implications of having done so). Then, you'll compute and examine summary statistics of these variables. This provides crucial context to the reader in terms of location and spread that will inform your analysis. Think of it this way: if you want to get a better understanding of how some variable Y varies with changes to some other variable X , you first need to know what typical values of Y and X are as well as how much they vary in general. Summary statistics provide a nice snapshot of this information. Another way to communicate this information is through data visualization. To this end, your data exploration should include some visualization of the distributions of your variables—most likely you'll want to use histograms for this.

Finally, you can move forward to analyzing your relationship of interest. As a first pass, you should compute summary statistics of your dependent variable grouped by some factors of your key explanatory variables. If your explanatory variable is continuous, you may need to create these groupings for the purpose of this analysis (one helpful way to do so is to group by whether observations are above or below the mean or median). Examine the differences in these summary statistics: does the center of the distribution vary a lot conditional on the explanatory variable? Does the spread? This leads very naturally into the next mode of analysis—t-tests. Is the difference between the means statistically significant? If you're a graduate student, you can take this analysis one step further by executing regression analysis. Of course, if you can't show it in a chart, you don't have a story. To this end, you'll complete your analysis by depicting your relationship of interest in a visualization. A great first step would be to plot histograms of your dependent variable grouped by the previously examined factors of your dependent variable. This will visually demonstrate the analysis you've already performed. You could also look at scatter plots that look at your relationship of interest, possibly averaging at levels of your dependent variable for simplicity. There's a lot of ways to get at this, and you should try many. Sometimes different charts may reveal new insights.

Criterion	Points Received	Points Possible
Introduction		10
Clearly introduce the research question (i.e., relationship of interest).		3
Provide a hypothesis describing this relationship.		4
Support the hypothesis with an intuitive explanation of the relevant mechanism.		3
Data Exploration		40
Cite the source of your data.		2
Explain the who/why/what/where/why/how of your data.		3
Identify and describe the key variables in the raw data including:		
special features such as outliers, top-coding, missing values etc.		
any transformations you performed to address these features		
how the variables connect to the relevant phenomena of your research		
any transformations you performed to better address your research question		10
Compute and interpret summary statistics of key variables.		10
Present and interpret descriptive visualizations of key variables.		15
Data Analysis		30
Compute and interpret grouped summary statistics.		10
Compute and interpret t-test(s).		10
Present and interpret comparative visualization.		10
Conclusion		10
Restate the research question and summarize findings.		4
Explain potential limitations of the research.		3
Identify avenues for future research.		3
Communication		10
External references are clearly cited.		
Spelling and grammar.		
Organization and flow.		
Clarity.		
Total		100